

Unit 3A

Transducers and sensors

Joan Oliver
14/10/2025

1

Biomedical Sensors and Signal Processing

1. Data acquisition systems and biomedical instrumentation concepts

2. Sensor data acquisition systems

3A. Transducers and sensors

- 3.1. Sensors and actuators introduction
- 3.2. Sensor classification and properties
- 3.3. Temperature sensors as a case example
- 3.4. Design optimization

3B. Transducers and sensors in biomedicine

- 3.5. Optoelectronic sensors.
- 3.6. Biosignals.
- 3.7. Biosensors.
- 3.8. Electrodes, electrocardiogram.
- 3.9. Medical image, X-rays.
- 3.10. Computerized tomography

4. Introduction to biomedical signals and linear systems Fourier analysis

5. Digital signal processing

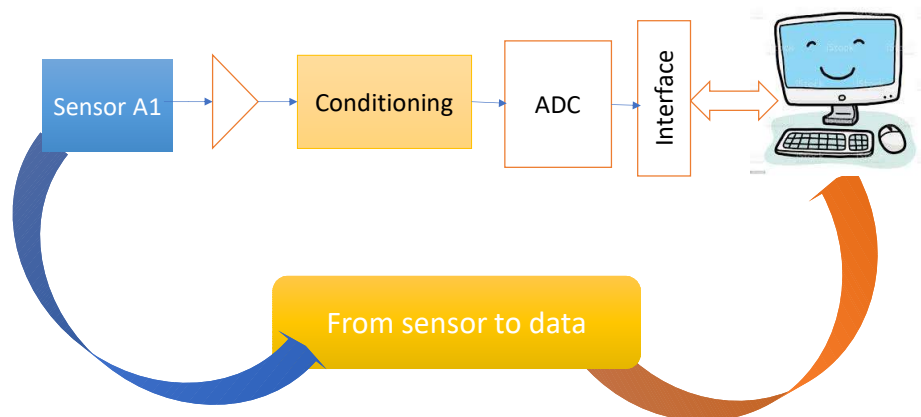
2

Bibliography

- M. Singh, *Introduction to Biomedical Instrumentation*, PHI Learning Private Limited, 2021.
- G. Webb, *Principles of Biomedical Instrumentation*, Cambridge University Press, 2018.

3

3.1. Sensors & actuators: ...in the big loop...



4

3.1. Sensors & actuators introduction

- **Transduction**

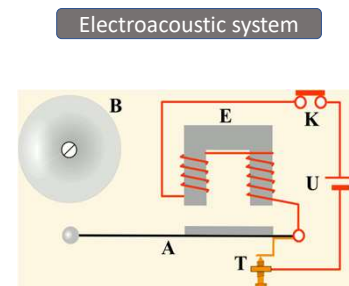
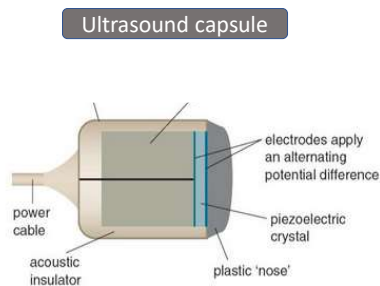
Principle relating energy transformation of a physical class of energy to another one

- **Transductor**

A transducer is a device that converts one type of physical quantity into another type of physical quantity.

A transducer consists of

- Exciting element/s
- Transducer element
- Packaging



3A. Transducers and sensors, Joan Oliver, 2025
Biomedical Sensors and Signal Processing

5

5

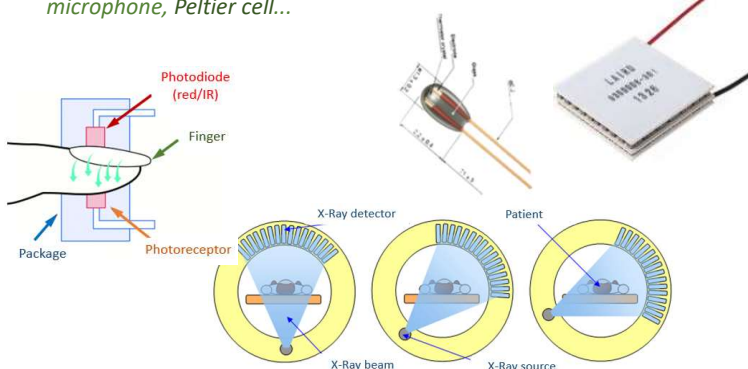
3.1. Sensors & actuators introduction

- **Sensor**

A sensor is a device sensitive to a physical quantity of the environment and transforms its energy, with a transducer, into a variation of an electrical variable.

Examples: (a lot)

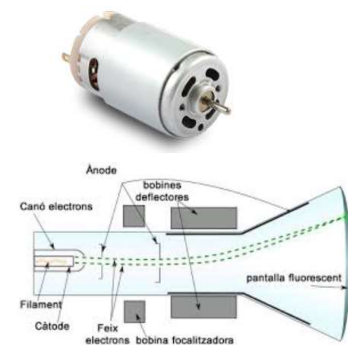
Thermistor, accelerometer, pulse oximeter, electrodes, microphone, Peltier cell...



- **Actuator**

An actuator transforms an electrical quantity into a physical one.

Examples: Cathode ray tube, motors, Peltier cell...



3A. Transducers and sensors, Joan Oliver, 2025
Biomedical Sensors and Signal Processing

6

6

3.1. Sensors, ...for multiple applications

- Temperature sensors
 - RTD
 - Termoparell
 - Termistor
 - Integrated temperature sensors
- Pressure sensors
- Optoelectric sensors
 - Light Dependent Resistor
 - Reflective, interrupts
 - Optocouplers
 - CMOS/CCD image sensors
 - Pulsioximeter
 - Photovoltaic cells
- Sound
 - Electret
 - Ultrasound
- Range sensors
 - Ultrasound
 - InfraRed
 - LIDAR
 - Range Laser Finders
- Inertial
 - Accelerometer
 - Gyroscope
- Inductive & magnètic
 - Efecte Hall
 - Inductiu: sensor velocitat
- Chemical sensors
- Biosensors
- Electrodes
- ETC.

7

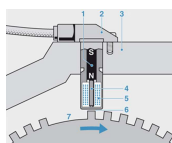
3.2. Sensors classification

Based on the output

- **Analogue:**
The output spans a continuous range of voltages. Signal conditioning usually requires amplification and digitization.
Example: most temperature sensors, microphone, pressure sensor, ...



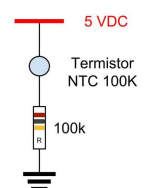
- **Digital:**
Sensors that have a discrete output voltage (logical 0 or 1). The output is usually encoded by a certain number of bits. They usually use a communication protocol (PWM, RS232, I2C or SPI, usually) to communicate with the processor.
Example: Encoder,...



8

Based on the power

- **Modulators (passive):**
They need the contribution of external energy.
Examples:
Thermistor: The quantity to be measured is a variation of the resistance.
Accelerometer: The quantity to be measured is a variation of the capacity.
Linear differential transformer: The quantity to be measured is a variation of the inductance, ...

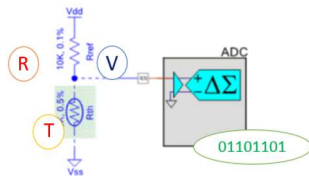


- **Self-generators (active):**
They can generate their own energy.
Examples:
Thermocouple, Peltier cell. They generate a potential difference depending on the temperature.
Piezoelectric. They generate a potential difference depending on a pressure differential.



3.2. Sensors classification: transduction mechanism

Thermistor (modulator)



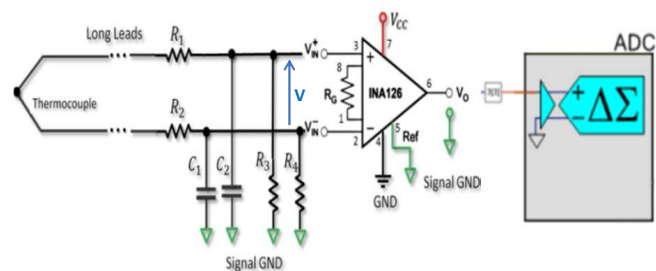
Transduction mechanism:

1. $T \rightarrow R$
2. $R \rightarrow V$
3. $V(\text{analog}) \rightarrow \text{Data}(\text{digital})$

Data reconstruction:

1. $\text{Data}(\text{digital}) \rightarrow V(\text{analog})$
2. $V \rightarrow R$
3. $R \rightarrow T$

Termocouple (self-generator)



3.2. Sensor properties

Static characteristics	Dynamic characteristics
Related with the response of the system under static input signals.	Response of the system when subjected to time-dependent signals Analysis is performed supposing a model of the system in: - First order. - Second order.
Precision and accuracy. Repeatability and reproducibility. Sensitivity. Range. Bias and drift. Linearity. Hysteresis. Systematic errors.	Dynamic error. Speed and response. Lag in the response.

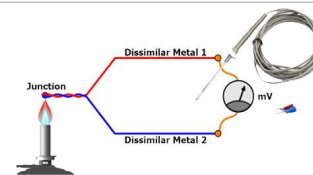
3.3. Temperature sensors as a case example

- Temperature sensors can be manufactured from several technologies that have really different properties.
 - RTD
 - Thermistors
 - Thermocouple
- These differences are paramount in selecting the best sensor for each application.

	RTD	Thermistor	Thermocouple
Temp. Range	😞	😞	😄
Stability	😄	😞	😞
Sensitivity	😞	😄	😞
Lineality	😄	😞	😞
System cost	😞	😄	😞
Applications	Highest accuracy	Best sensitivity	Highest temp.

3.3. Temperature sensors: Thermocouple

- It consists of two wires of different metals welded together. At the other end of the wires is another junction (reference junction), electronically compensated for ambient temperature.
- When the sensory junction is heated, a thermoelectric potential of ΔV is generated proportional to the temperature difference between the two junctions.
- It bases on the physical working principle of the Seebeck effect: two wires joined together forming a loop with a metal junction, causes a current to flow through the loop.
- Conditioning requires amplification.



Model:

Approximated by a polynomial where the coefficients are obtained from reference temperatures (e.g., 0°C and 100°C)

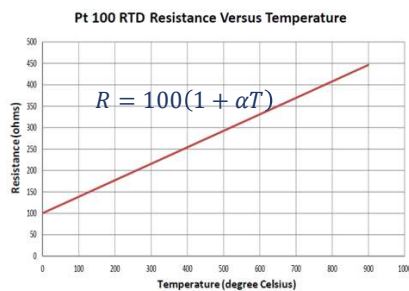
$$V(T) = a_1 \cdot T + a_2 \cdot T^2 + a_3 \cdot T^3 + \dots + a_8 \cdot T^8$$

Type	Positive lead	Negative lead	Temperature range	Temperature coeff. variation $\mu V/^\circ C$	Most linear range and sensitivity in the range
R	Platinum-Rhodium (87% Pt, 13% Rh)	Platinum	0-1500°C	5.25-14.1	1100-1500°C 13.6-14.1 $\mu V/^\circ C$
S	Platinum-Rhodium (90% Pt, 10% Rh)	Platinum	0-1500°C	5.4-12.2	1100-1500°C 13.6-14.1 $\mu V/^\circ C$
K	Chromel (90% Ni, 10% Cr)	Alumel (Ni ₉₀ Al ₁₀ MnSi)	-200-1300°C	15.2-42.6	0-1000°C 38-42.9 $\mu V/^\circ C$
E	Chromel	Constantan (57% Cu, 43% Ni)	-200-1000°C	25.1-80.8	300-800°C 77.9-80.8 $\mu V/^\circ C$
T	Copper	Constantan	-200-350°C	15.8-61.8	nonlinear
J	Iron	Constantan	-150-750°C	21.8-64.6	100-500°C 54.4-55.9

3.3. Temperature sensors: Resistance Temperature Detector (RTD)

- Consists of a wire coil or deposited film of pure metal.
- The element's resistance increases with temperature in a known and repeatable manner.
- Pt100 RTD model:

$$R = R_0(1 + \alpha(T - T_0))$$



Material	Temperature Coefficient of Resistance - α - ($1/^\circ\text{C}$, $\Omega/\Omega^\circ\text{C}$)
Nickel	0.0067
Iron	0.002 to 0.006
Tungsten	0.0048
Aluminum	0.0045
Copper	0.0043
Lead	0.0042
Silver	0.0041
Gold	0.004
Platinum	0.00392
Mercury	0.0009
Manganin	+/- 0.00002
Carbon	-0.0007
Electrolytes	-0.02 to -0.09
Thermistor	-0.068 to 0.14

3A. Transducers and sensors, Joan Oliver, 2025
Biomedical Sensors and Signal Processing

13

13

3.3. Temperature sensors: Thermistor

- Resistive device made of metal oxides soldered in a ball of epoxy or glass.
- The principle of operation is a non-linear resistance variation with temperature. They can be of **positive (PTC)** or **negative (NTC)** coefficient.
- They present a smaller temperature range than RTDs, but they are much more sensitive.
- The thermistor resistance at 25 °C is taken as *base resistance*. Thus, the plot on the right is of a 30 kΩ NTC.
- Lineal model is based on the **Temperature Coefficient α** . It is the material characteristic given in percentage of resistance change per degree Centigrade, being R_T the resistance at a relevant T and dR/dT the gradient at T.

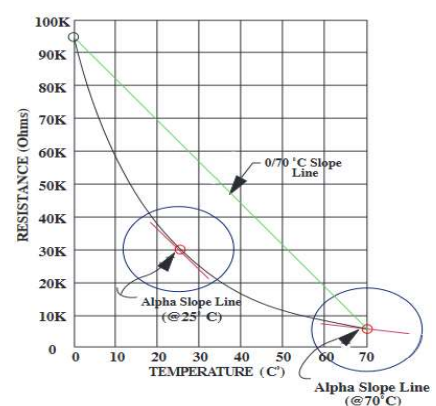
$$\alpha = \frac{1}{R_T} \frac{dR}{dT} 100 \left(\frac{\%}{^\circ\text{C}} \right)$$

Example: A thermistor has a resistance of 30 kΩ at 25 °C with $\alpha = -4.39 \text{ } \%/^\circ\text{C}$.

A new resistance measurement gives 31 kΩ. What temperature is the device at?

$$\Delta T = \frac{\Delta R}{R_T} \cdot \frac{100}{\alpha} = \frac{R_T - R_{25^\circ}}{R_{25^\circ}} \cdot \frac{100}{\alpha} = \frac{31 - 30}{30} \cdot \frac{100}{-4.39} = -0.76 \text{ } ^\circ\text{C}$$

$$\Delta T = T_T - T_{25} \Rightarrow T_T = \Delta T + T_{25} = 24.24 \text{ } ^\circ\text{C}$$



3A. Transducers and sensors, Joan Oliver, 2025
Biomedical Sensors and Signal Processing

14

14

3.3. Temperature sensors: Thermistor (II)

- One parameter or *α model* (linear)
 - α parameter is given at spaced temperatures.
 - Only suitable for small temperature variations.
- 2 parameters or *β model* (exponential)
 - α calculated in the datasheet at two specific temperatures
 - When temperature spans for more that 40°, β model becomes not so precise.
- 3 parameter or *Steinhart-Hart model*
 - Fits better the thermistor behavior
 - Given the resistance at three temperatures, the parameters are calculated solving the corresponding 3-equation system with 3 unknowns.

$$\alpha = \frac{1}{R_T} \cdot \frac{dR}{dT} \cdot 100 \quad (\%/^{\circ}C)$$

$$R_T = R_{T_0} e^{\beta \left(\frac{1}{T} - \frac{1}{T_0} \right)}$$

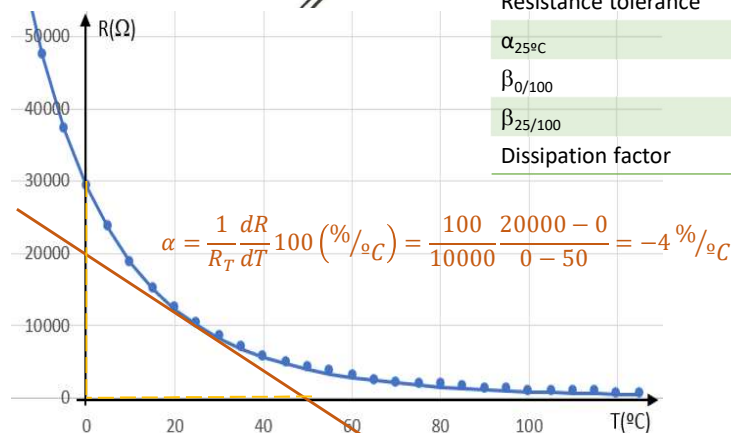
$$\frac{1}{T} = A + B \cdot \ln R + C \cdot (\ln R)^3$$

15

3.3. Thermistor case example (I)

- **Example 1.** Plot the B57551G1103F005 datasheet table.
 - From the plot find the α parameter at 25 °C

T (°C)	R (Ω)
0	29287
5	23311
10	18684
15	15075
20	12240
25	10000
30	8217
35	6790
40	5642
45	4712
50	3955
55	3335
60	2826
65	2404
70	2055
75	1763
80	1519
85	1314
90	1140
95	993
100	868



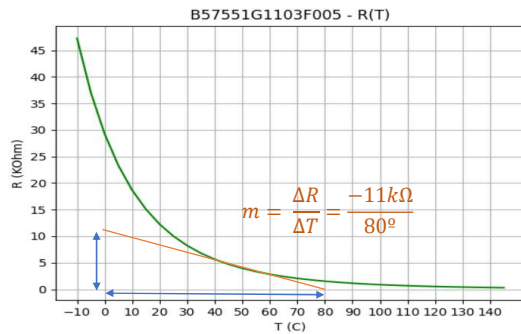
From datasheet

R25	10 kOhm
Resistance tolerance	±1%
α _{25°C}	-4 %/°C
β _{0/100}	3586 K
β _{25/100}	3625 K
Dissipation factor	0.8 mW/K

16

3.3. Thermistor case example (IV)

- Example 2.** Using $\alpha_{50^\circ\text{C}}$ parameter of the thermistor, find the error introduced using the linear model. Consider the resistance of the thermistor from 40 °C to 60 °C, in steps of 2.5 °.



1 $\alpha_{50^\circ\text{C}}$ is the slope of the graph at 50 °C. Then:

$$\alpha_{50^\circ\text{C}} = \frac{1}{R_T} \cdot \frac{dR}{dT} \cdot 100 = \frac{100}{3955} \cdot \frac{0-110}{80-0} \cong -3.5 \text{ }^\circ\text{C}^{-1}$$

$$\alpha_{50^\circ\text{C}}(\text{datasheet}) = -3.5 \text{ }^\circ\text{C}^{-1}$$

T (°C)	R (Ω)	R _{datasheet} (Ω)	Error (%)
40	5339	5642	5.4
42.5	4993		
45	4647	4712	1.4
47.5	4301		
50	3955	3955	0
52.5	3609		
55	3363	3336	2.2
57.5	2917		
60	2570	2405	9

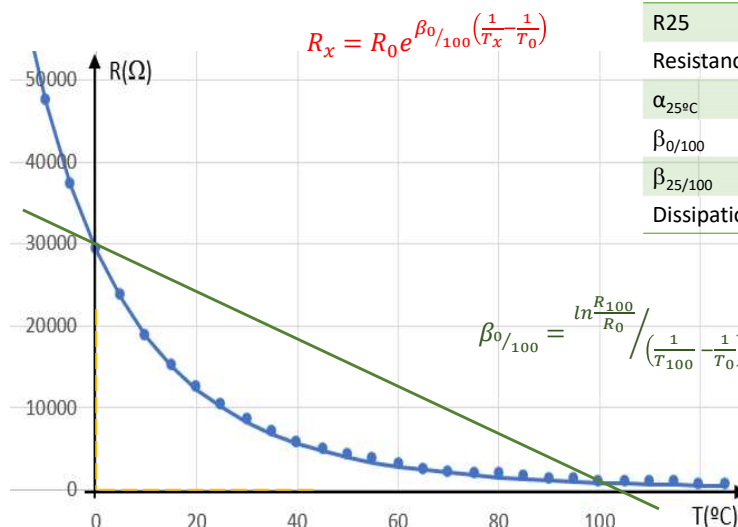
2 Working with $\alpha_{50^\circ\text{C}}$ equation:

$$R_x = R_{50^\circ\text{C}} + \alpha_{50^\circ\text{C}} R_{50^\circ\text{C}} (T_x - T_{50^\circ\text{C}}) / 100$$

$$R_x = 3955 + (-3.5 \cdot 3955 (T_x - 50) / 100)$$

3.3. Thermistor case example (II)

- Example 3.** Taking table R/T data, find the $\beta_{0/100}$ parameter



From datasheet

R25	10 kOhm
Resistance tolerance	±1%
$\alpha_{25^\circ\text{C}}$	-4 %/°C
$\beta_{0/100}$	3586 K
$\beta_{25/100}$	3625 K
Dissipation factor	0.8 mW/K

3.3. Thermistor case example (III)

- Example 4.** Using the Steinhart-Hart model of the thermistor, find the resistance of the thermistor at 50 °C with parameters taken at 0 °C, 80 °C i 200 °C.

Step 1: Calculate coefficients

Applying equation $\frac{1}{T} = A + B \cdot \ln R + C \cdot (\ln R)^3$ at 0 °C, 80 °C i 200 °C:

$$\begin{cases} \frac{1}{273.15} = A + B \cdot \ln 29287 + C \cdot (\ln 29287)^3 \\ \frac{1}{353.15} = A + B \cdot \ln 1519.5 + C \cdot (\ln 1519.5)^3 \\ \frac{1}{473.15} = A + B \cdot \ln 103.38 + C \cdot (\ln 103.38)^3 \end{cases} \quad \begin{cases} A = 9.16489353e-04 \\ B = 2.558208e-04 \\ C = 1.04245596e-07 \end{cases}$$

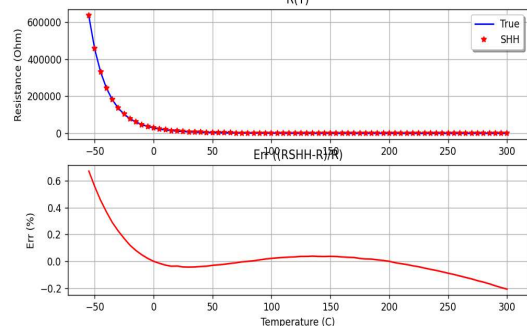
Step 2: Calculate the resistance at 50 °C

$$\frac{1}{T} = 9.16489353e-04 + 2.558208e-04 \cdot \ln R_{50} + 1.04245596e-07 \cdot (\ln R_{50})^3$$

And solving the equation (remember manufacturer $R_{50°C}=3955 \Omega$)

$$R = 3953.82 \Omega$$

$$\text{Error} = 0.03\%$$



Inverse Steinhart-Hart

$$R = \exp\left(\sqrt[3]{y - x/2} - \sqrt[3]{y + x/2}\right)$$

$$x = \frac{1}{C} \left(A - \frac{1}{T} \right)$$

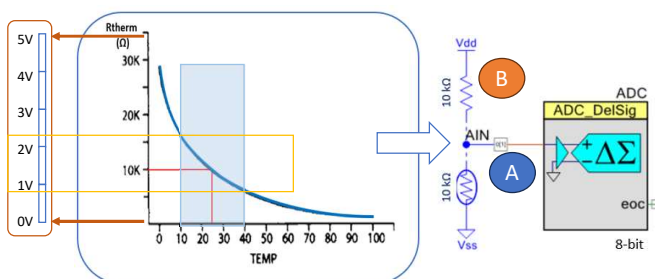
$$y = \sqrt{\left(\frac{B}{3C}\right)^3 + \frac{x^2}{4}}$$

3.4. Design optimization: maximizing the ADC input span

- Sensor interface building could be cumbersome, and usually needs of adhoc approaching
- Consider the case of a thermistor. It presents two challenges:
 - Maximize the scaling of the output → **A**
 - Find the best load resistance to minimize self heating → **B**

A first design approach

Full output (resistance) range



Actual data range:

- At 40 °C → $V_{AIN} = 1.8 \text{ V}$
- At 10 °C → $V_{AIN} = 3.25 \text{ V}$

$$40^\circ \rightarrow 1.8\text{V} \cdot \frac{256\text{steps}}{5\text{V}} = 92\text{steps}$$

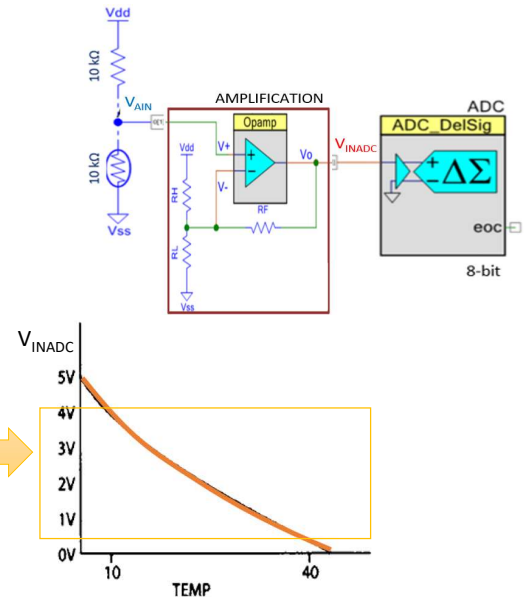
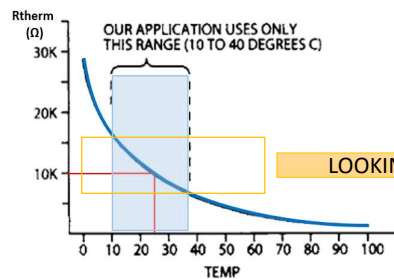
$$10^\circ \rightarrow 3.25\text{V} \cdot \frac{256\text{steps}}{5\text{V}} = 166\text{steps}$$

$$\left| \frac{92-16}{40-10} \right| = 2.47 \text{ ADCsteps}/^\circ \rightarrow 0.405^\circ/\text{step}$$

3.4. Design optimization: maximum input span (I)

System requirements

- Working T_{range} from 10 °C to 40 °C
- Maximize output span
- (try) Accuracy of at least 0.2 °C/step
- 8-bit ADC



3A. Transducers and sensors, Joan Oliver, 2025
Biomedical Sensors and Signal Processing

21

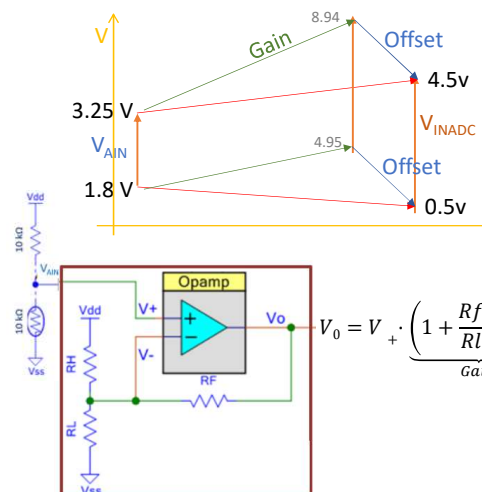
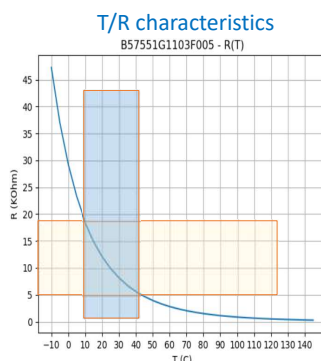
21

3.4. Design optimization: maximum input span (III)

Second approach

- It is necessary to shift the analog signal to put it in the correct range for the A/D converter.

T (°C)	R (Ω)
0	29287
5	23311
10	18684
15	15075
20	12240
25	10000
30	8217
35	6790
40	5642
45	4712
50	3955
55	3335
60	2826
65	2404
70	2055
75	1763
80	1519
85	1314
90	1140
95	993
100	868



$$V_0 = V_{AIN} \cdot \left(1 + \frac{R_f}{R_l} + \frac{R_f}{R_h} \right) - \frac{V_r R_f}{R_h} \text{ Offset}$$

3A. Transducers and sensors, Joan Oliver, 2025
Biomedical Sensors and Signal Processing

22

22

3.4. Design optimization: maximum input span (IV)

- Let's a range span from 0.5V to 4.5V

$$gain = \frac{4.5 - 0.5}{3.25 - 1.8} = 2.75 = 1 + \frac{Rf}{Rl} + \frac{Rf}{Rh}$$

- Introducing the gain, outputs are:

$$\begin{aligned} 10^\circ &\rightarrow 3.25 \cdot 2.75 = 8.9V \\ 40^\circ &\rightarrow 1.8 \cdot 2.75 = 4.9V \end{aligned} \rightarrow span = 4V$$

- Offset (from each of these two voltages) is (must be the same):

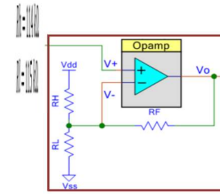
$$8.9 - 4.5 = 4.4V = \frac{Vr \cdot Rf}{Rh}$$

$$4.9 - 0.5 = 4.4V = \frac{Vr \cdot Rf}{Rh}$$

- Solving gain and offset equations and selecting $Rf=100\text{ k}\Omega$ ($Vr = 5V$):

$$Rh = 114\text{ k}\Omega$$

$$Rl = 115\text{ k}\Omega$$



New data range:

- At $40^\circ\text{C} \rightarrow V_{INADC} = 0.5\text{ V}$
- At $10^\circ\text{C} \rightarrow V_{INADC} = 4.5\text{ V}$

$$\begin{aligned} 40^\circ &\rightarrow 0.5V \cdot \frac{256\text{steps}}{5V} = 26\text{steps} \\ 10^\circ &\rightarrow 4.5V \cdot \frac{256\text{steps}}{5V} = 230\text{steps} \\ \left| \frac{26-2}{40-10} \right| &= 6.8\text{ ADCsteps}/^\circ \rightarrow 0.15^\circ/\text{step} \end{aligned}$$

3.4. Design optimization: self-heating (I)

Self-heating

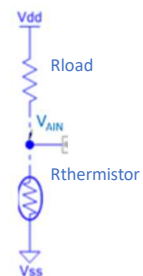
- Thermistor $\rightarrow$ Resistor $\rightarrow$ A heating device
A large R_{load} minimizes self-heating
- Dissipation constant (D.C.) $\rightarrow$ Number of milliwatts needed to raise the thermistor temperature 1° above ambient.
- Units are W/K.
- The amount of self-heating depends on the thermistor accuracy. A $\pm 5^\circ$ tolerance thermistor tolerates more thermistor self-heating than a $\pm 1^\circ$ tolerance thermistor.
- Lets take that the power dissipation allowed is:

$$P = D.C. * RequiredAccuracy(in^\circ\text{C})$$

(Example, the B57551G1103F005 $\rightarrow$ D.C.: 0.8mW/K)

- So, the dissipation power (in terms of D.C.) is $P = D.C. * T_{increase} \rightarrow T_{increase} = \frac{P}{D.C.} = \frac{V^2}{D.C.*R}$
- Then, the difference between ambient temperature and measured temperature can be obtained in terms of self-heating.

$$\text{Self-heating} = T(\text{measured by the thermistor}) - \frac{V^2}{D.C.*R}$$



3.4. Design optimization: self-heating (II)

- **Case 1.** Find the allowed thermal dissipation (in this thermistor) if it is required an accuracy of 1°.

Thermal dissipation = D.C. * (Wanted Accuracy) = 0.8mW/K*1K = 0.8mW

- **Case 2.** Which is the limit of R_{pullup} (in the scaling example) in order to maintain power dissipation within limits? That is, self-heating <1° C, or, thermal dissipation < 0.8mW

The maximum current passing through the thermistor must be calculated. It occurs at...

The minimum thermistor resistance (at 40°) = 5.642 KΩ

Thermistor voltage drop at this resistance:

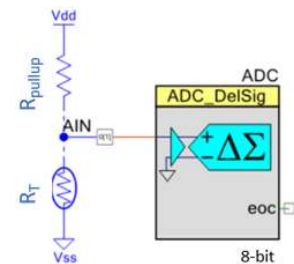
$$P = 0.8 = \frac{V^2}{R} \rightarrow V = \sqrt{0.0008 * 5642} = 2.124V$$

Current through the thermistor:

$$I = \frac{2.124 - 0}{5642} = 0.000376A$$

Resistor minimum value:

$$R_{pullup} = \frac{V_{ref} - 2.124}{0.000376} = 7.65K\Omega$$



Biomedical Sensors and Signal Processing

